Thakaa at AraGenre 2026: Definition-Guided LLM Judging with Full-Context Multi-Model Consensus for Hierarchical Arabic Genre Classification

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Abstract

We describe the winning system for the AraGenre 2026 shared task on hierarchical, definition-guided Arabic genre classification, where each of 27,972 hidden-test Arabic segments must be assigned one of 6 broad genres and one of 74 fine-grained specific genres whose English definitions are provided but whose label set is *disjoint* from the released training data (fully zero-shot). Our approach uses no task training. A **family-restricted, definition-guided LLM judge** performs set-wise scoring of candidate genres within a hierarchically constrained decoding scheme, reaching 0.7139 Hierarchical Macro F1. From the public scoreboard we diagnose that our system already led on Specific Macro F1 but trailed on Broad Macro F1, and we close that gap with a **full-context multi-model consensus** correction stage: two instruction models are each shown the *entire* 74-definition taxonomy and re-predict the specific genre; corrections are applied only where both models agree against the base system. Feeding the full taxonomy makes the models far more resistant to a topic trap (confusing a book *about* religion with religious scripture). A final attractor-drain step redistributes over-predicted generic classes to starved rare classes. The system scores **0.7352 Hierarchical Macro F1, ranking 1st**, $+0.018$ over the next team and improving every metric over our base.

1 Introduction

AraGenre 2026 (El-Haj et al., 2026) frames Arabic genre classification as a generalisation problem: systems receive English genre definitions and limited synthetic training data, but the hidden benchmark contains noisier naturally occurring Arabic spanning Modern Standard Arabic, Classical Arabic, and multiple dialects, with *previously unseen genre–definition combinations*. Because the 74 specific test genres are disjoint from training, memorisation is useless and semantic understanding of communicative function is required. The official ranking metric is

$$\text{Hier. F1} = \tfrac{1}{2}\left(\text{Broad F1}_{\text{macro}} + \text{Spec. F1}_{\text{macro}}\right).$$

Our contributions are:

1. A hierarchically constrained, definition-guided LLM judge that reaches 0.7139 Hierarchical Macro F1 with no task training.
2. A scoreboard-driven diagnosis showing the two tiers must be optimised differently, and that our residual error was concentrated in the broad tier.
3. A full-context multi-model consensus correction, applied conservatively (agreement-gated, direction-verified), with a three-arm ablation separating the effect of the taxonomy from that of an explicit type-vs-topic instruction.
4. An attractor-drain step targeting the macro-F1 failure mode of over-predicting generic classes.
5. A winning result (0.7352 Hierarchical Macro F1, rank 1 of all participating teams), with each component scored as a submission.

Code and prompts: https://github.com/Hassn11q/thakaa-aragenre-2026. The main challenges were a fully disjoint label set, a topic trap (writing *about* a subject vs. the subject's genre), and classes with no written signal (song dialects, textbook levels).

2 Related Work

Arabic and hierarchy-aware HTC. Arabic pretrained models (Antoun et al., 2020, 2021) underpin Arabic classification, including hierarchical variants (Bouchiha et al., 2025, 2026), and a large literature models the label hierarchy explicitly (Zhu et al., 2023; Cai et al., 2024; Zeng et al., 2025; Wang et al., 2025; du Toit and Dunaiski, 2025; Zangari et al., 2024). These closed-label supervised setups do not transfer to AraGenre's unseen, definition-specified labels; we instead enforce the hierarchy structurally, forcing broad $=$ parent(specific).

Zero-shot HTC, LLM judging, and consensus. Zero-shot HTC matches text to label semantics rather than fixed classes; prompt-ensembling (Xia et al., 2025), LLM-enhanced prototypes (Zhang et al., 2025), LLM-refined taxonomies (Golde et al., 2026), consistency regularisation (Pan and Wu, 2025) and reasoning-based HTC (Jiang et al., 2025) inform our definition refinement and chain-of-thought pass. Our two-stage design retrieves candidate definitions (Chen et al., 2024) then compares them with an LLM (Sun et al., 2023; Zheng et al., 2023), using Qwen3-Embedding (Qwen Team, 2025) to bridge Arabic text and English definitions. Combining independent verdicts builds on ensembling (Dietterich, 2000) and its LLM forms: juries (Verga et al., 2024), blending (Jiang et al., 2023), mixtures of agents (Wang et al., 2024), debate (Du et al., 2024), and self-consistency (Wang et al., 2023); diverse ensembles also generalise better to unseen distributions. Unlike juries voting on free-form text, we require *exact-label* agreement, and find it reliable only with the full taxonomy in context.

3 Background

Each instance is `{id, text}`; systems output one `broad_genre` (of 6: Informative, Creative, Interactive, Learning, Legal, Religious) and one `specific_genre` (of 74), using exact taxonomy label names, each with an English definition. The released data are tiny and synthetic-like (train: 140 items / 7 specific genres; dev: 110 / 6), and the specific-label inventories are *100% disjoint* across train, dev and test, so no supervised label set transfers. The *hidden test* has 27,972 naturally occurring instances over 74 specific genres in 6 families: Informative (42, dominated by 20 job domains and 13 book-description topics), Learning (12: textbooks by level, student writing by topic), Creative (7: 5 song dialects, classical vs. MSA poetry), Religious (6: Islamic and Christian), Interactive (4) and Legal (3). Missing predictions are counted incorrect; official dev baselines are in Table 1. Informative dominates (~50% of predictions), and the specific tier's 74 classes make Macro F1 highly sensitive to rare-class recall (e.g., dialect song lyrics, niche job domains).

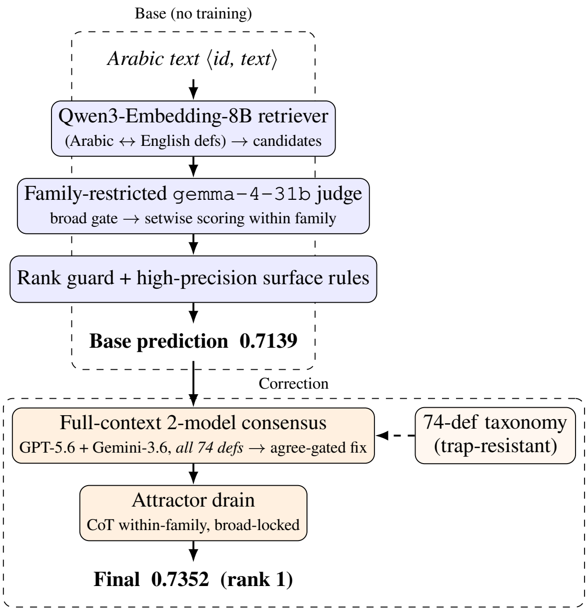

Figure 1: System overview. The base judge yields 0.7139; full-context two-model consensus plus a within-family attractor drain correct both tiers, reaching 0.7352 (rank 1).

4 System

4.1 Base: Family-Restricted Definition-Guided Judge

A locally served open Gemma-family model (Gemma Team, Google DeepMind, 2024), exposed by our vLLM endpoint under the identifier `gemma-4-31b`, acts as a setwise judge; we report the served identifier verbatim and pin it, together with the API model strings, in Appendix G. For each text: (i) `Qwen3-Embedding-8B` (Qwen Team, 2025) encodes the Arabic text and the English candidate definitions and proposes candidate specific genres by cosine similarity, bridging Arabic text and English definitions; (ii) decoding is hierarchically constrained: the broad genre is gated first and specific candidates are restricted to that broad family (all siblings for small families, top-k retrieved for large topical ones), so that broad $=$ parent(specific) and broad *accuracy* equals the family-accuracy of the specific prediction (Broad Macro F1 does not: it averages per-class F1, and the gap between the two is what §4.2 exploits); (iii) the judge scores each candidate 0–100 against its definition and we take the argmax, with averaging over candidate orderings to counter listwise position bias (Sun et al., 2023) and light marginal calibration; (iv) a conservative embedding-rank guard reverts low-confidence judge overrides; and (v) a few high-precision surface rules override the judge only on near-unambiguous markers (Qur'anic mushaf orthography → `quran`; isnād openings → `hadith`; both estimated to fire at >98% precision by inspecting a sample of firings, as the test split has no gold; emoji/hashtag short posts → Interactive subtypes). We also supply 74 offline-generated per-genre contrastive discriminators; hyperparameters are in Appendix G. On the hidden test this base pipeline scores **0.7139**.

4.2 Diagnosis from the Scoreboard

The leaderboard exposed the structure (Table 2): our base led on Specific Macro F1 (0.5487 vs. 0.5339) but trailed on Broad Macro F1 (0.8792 vs. 0.90). As the ranking metric is their mean, the entire deficit was in the broad tier. Broad accuracy (0.9003) exceeding broad macro (0.8792) indicated a minority broad class dragged the mean via cross-family false positives: Bible passages filed as literature, news filed as legal, informal comments filed as book descriptions.

4.3 Full-Context Multi-Model Consensus Correction

We re-predict the specific genre for suspect instances with two frontier instruction models (GPT-5.6 Luna (OpenAI, 2026) and Gemini-3.6 Flash (Google, 2026)), *feeding each the entire 74-definition taxonomy* (≈5.7K tokens) grouped by broad genre, and asking for the single best specific label. This follows the classical ensemble argument that combining diverse, independently-erring predictors beats any single predictor (Dietterich, 2000), in its LLM form (Verga et al., 2024; Wang et al., 2024). (A third model, Claude Sonnet 5 (Anthropic, 2026), was piloted but excluded on cost grounds.)

What suppresses the topic trap. With only the 6 broad definitions, all models fell into a *topic trap*: text *about* Islam → Religious, even when it was an Informative *book description*. The full-taxonomy prompt also carries an explicit type-vs-topic instruction, so we ran a three-arm ablation over the same 1,403 book-description instances to separate the two factors (one model, GPT-5.6, identical pool): with broad definitions only and no such instruction, 394 are called Religious; adding the instruction alone removes 41% of those (394→233); supplying the full taxonomy removes a further 46% (233→126). Both factors matter and neither explains the effect alone: the instruction states the convention, the taxonomy shows the label that realises it. Scripture misfiled as fiction (هؤلاء امراء بني عيسو, Genesis 36) still moves to Religious, and base-family agreement rises in four of the six families (Appendix D).

Conservative application. A correction is applied only where both models agree on the same specific label differing from the base. Broad-changing moves are gated to verified-safe directions (Legal→Informative for wire news, Informative→Interactive for comments, Creative/Informative→Religious for scripture); read-verified-unreliable directions (e.g., Informative→Learning, which over-fires student-writing on academic prose) are rejected. Within-family (broad-preserving) refinements are applied wherever both models agree.

4.4 Attractor Drain (Specific Macro)

Manual reading revealed five large "attractor" classes (e.g., `egyptian_song_lyrics`) that absorb texts the judge cannot confidently place, inflating those classes and starving rare siblings. The drain is deterministic: when the base label is an attractor class and an independent chain-of-thought re-judge (Wei et al., 2022) names a different sibling in the same family, the sibling replaces it (broad never changes). On the base alone it moves 554 instances, 6–35% of each attractor class (Appendix E); in the final pipeline consensus runs first, leaving the drain 383. Relative to the base, the final system changes 459 broad and 2,890 specific labels (10.3% of instances).

5 Experimental Setup

Data splits. We train nothing; every system is applied unchanged to the hidden test. An input is `{"id","text"}` with Arabic text such as مطلوب سائق لديه رخصة قيادة ("driver wanted, must hold a licence"); the required output pairs it with `broad_genre` = Informative and `specific_genre` = `drivers_and_delivery_jobs`.

Preprocessing. None. Normalisation erases the two orthographic signals that separate whole families here: heavy diacritisation marks 77% of the

System	Split	Hier. F1
Baseline 1 (embedding similarity)	dev	0.4658
Baseline 2 (RAG + multilingual LLM)	dev	0.3899
Our judge (`gemma`)	dev	0.8794
Our judge (GPT-5.6)	dev	0.8943
+ consensus	dev	0.8557
GPT-5.6 judge + attractor drain†	dev	0.9358
Genre-general zero-shot judge	test	0.6067
+ embedding retrieval gate	test	0.6812
family-restricted, derived broad	test	0.6882
+ rules / calibration (base)	test	**0.7139**
+ 2-model consensus (pool)	test	0.7224
+ safe broad expansion (full set)	test	0.7256
+ full consensus + drain (final)	test	**0.7352**

Table 1: Main progression on both splits. Dev is near-degenerate (its 6 specific genres map one-to-one onto 6 broad, so all three F1 metrics coincide), so the hidden test is our primary evaluation. †applied to the GPT-5.6 judge, tuned on dev, so not held out.

texts we label `quran` and 43% of poetry against 1% elsewhere, and emoji mark 47% of Interactive posts against under 1% (Appendix G).

Tools. `transformers`, `sentence-transformers` and `bitsandbytes` for Qwen3-Embedding-8B, vLLM for the local judge, and the `openai` and `google-genai` SDKs for the verifiers; versions and URLs in Appendix G.

Metrics. Macro F1, Weighted F1 and Accuracy at both tiers; the official ranking metric is Hierarchical Macro F1, the mean of Broad and Specific Macro F1.

Cost. Consensus sends a ≈6.2K-token prompt per instance, about US$1.05–1.48 per 1,000 instances and $50 in total; wall-clock timings are in Appendix G. The base pipeline needs no paid API.

6 Results

6.1 Development-Set Results and Component Transfer

Re-running the pipeline on the development set (110 examples, official gold) separates what transfers from what does not. The drain transfers: its attractor classes are detected from the prediction distribution alone (`Analytical Reports` 1.40× its gold frequency), with no class list carried over, and it adds +0.042. The consensus stage does not: dev has no `*_book_description` and no scripture classes, so the trap it repairs has zero instances, only two corrections fire, and both are wrong. Residual dev errors stem from a convention clash between splits (Appendix A).

System	Hier	SpM	SpW	SpA	BrM	BrW	BrA
Best rival‡	.717	.534	.581	.599	**.900**	.912	.911
Base (ours)	.714	.549	.584	.591	.879	.900	.900
+ consensus	.722	.553	.592	.598	.892	.910	.910
Final (rank 1)	**.735**	**.573**	**.613**	**.619**	.898	**.914**	**.914**

Table 2: Official metrics on the hidden test, rounded to three decimals (Sp/Br = Specific/Broad; M/W/A = Macro F1, Weighted F1, Accuracy). We beat the prior #1 on six of seven; the exception is Broad Macro F1 (.898 vs. .900), and we lead Specific Macro F1 by +.039. Rank is from the final official standings; ‡is the strongest rival on the public leaderboard, snapshot of 30 July 2026.

7 Analysis

Broad Macro is dominated by minority-class false positives and Specific Macro by rare-class recall, so small verified gains on both compound through the mean. Broad-only prompting produced 180 unanimous topic-trap errors even across all three models we tried, which the full taxonomy dissolved; and any single strong model disagreed with the base on ∼29% of specifics, too noisy to trust, whereas two-model exact-label agreement filtered this to reliable corrections. A direct six-way broad classifier regressed to 0.6206 and unpruned 74-way scoring to 0.6517; self-consistency (0.7134), optimal transport (0.7115) and uniform-prior calibration (0.7124) all fell below the 0.7139 base (Appendix C).

Error analysis (hidden test). Three types dominate: cross-family topic leaks (a description *of* a religious book attracting Religious, and scripture misfiled as fiction), attractor over-prediction, and ceiling classes where 87–99% of the texts we label as song lyrics carry no marker from that dialect's own cue list. Examples in Appendix B.

8 Conclusion

Zero-shot hierarchical Arabic genre classification is achievable without task training: a constrained definition-guided judge, an agreement-gated consensus and a rare-class attractor drain ranked 1st at 0.7352 Hierarchical Macro F1. Our ablation shows the topic trap is suppressed jointly by a type-vs-topic instruction and by the full label inventory, neither alone. The remaining ceilings are song-lyric dialect and textbook reading level.

Acknowledgements

We thank the AraGenre 2026 organisers for designing and running the shared task, for the released taxonomy and definitions, and for their prompt support throughout the evaluation phase. We also thank the maintainers of the open models and libraries our pipeline builds on. Our code, prompts, cached model outputs and submission builder will be released at https://github.com/Hassn11q/thakaa-aragenre-2026 upon publication.

Limitations

All reported numbers are official evaluation-phase submissions; we ran none post hoc. The hidden test has no released gold labels and the development set is near-degenerate, so component decisions were validated on the public leaderboard and by manual reading rather than on held-out gold. Our reported progression therefore doubles as the selection signal: successive submissions were guided by leaderboard feedback, which risks fitting to the test distribution, and we report no held-out confidence intervals or significance tests. The gains should be read as leaderboard-measured rather than statistically certified, and, absent gold, we can offer no uncertainty estimate on the winning margin. Consensus correction incurs API cost proportional to test size. Song-dialect distinctions remain near the noise floor because the distinguishing features are phonetic and largely absent from written lyrics.

Reproducibility Statement

Official submission. The results we report are those of submission 872515, evaluated 1 August 2026, which scored 0.7352 Hierarchical Macro F1 (Specific Macro 0.5725, Weighted 0.6125, Accuracy 0.6189; Broad Macro 0.8979, Weighted 0.9142, Accuracy 0.9138) over 27,972 of 27,972 gold instances with no missing prediction. Every hidden-test number in Table 1 and Appendix C is likewise an official scored submission rather than a local estimate; the submission IDs are listed in Appendix H. All inference is zero-shot with no task-specific training. **Data:** we use only the official AraGenre inputs and definition files; the full 74-entry `test_genre_definitions.json` is fed to every model verbatim. Three surface rules and one prompt clause do name specific labels (`quran`, `hadith`, `*_book_description`); nothing else is hard-coded. **Models:** the base retriever is Qwen3-Embedding-8B (4-bit NF4 weights with fp16 compute, 4096-dim, cosine over L2-normalised vectors, sequence length capped at 512 tokens); the judge is a Gemma-family model (`gemma-4-31b`) served via vLLM with setwise 0–100 scoring; the consensus verifiers are GPT-5.6 Luna and Gemini-3.6 Flash, each queried with the full-taxonomy prompt. **Settings:** decoding temperature 0 with a fixed seed throughout; retrieval k=12 narrowed to $\leq$6 family-restricted candidates; permutation self-consistency K=3 on confusable families; marginal calibration α=0.5 with SLD rare-class correction; embedding-rank guard at rank 3. **Pipeline and code:** the retriever, judge, full-context consensus verifiers (`src/verify.py`), chain-of-thought drain, and the submission builder (`src/submission.py`, which enforces broad=parent(specific) and validates that all 27,972 IDs are present with exact labels) are released in the project repository. **Two cached inputs.** The broad gate the judge restricts to is submission 866538 (0.6812), an earlier retrieval-gated system we submitted and scored, reused verbatim rather than regenerated by the released code; and the base predictions additionally carry an Interactive-recall reassignment applied during the evaluation phase that was never preserved as a script; re-running the released judge over the same candidate sets reproduces 93.5% of the base specific labels and 97.1% of the broad labels. Both files ship in `artifacts/`, so the submitted predictions rebuild exactly (27,972/27,972) from cached outputs, but the base stage is a cached input rather than something the repository regenerates end to end. **Caveats:** the hosted API models are versioned by release date and are not guaranteed bit-reproducible across dates; the consensus stage incurs API cost proportional to the number of verified instances (on the order of tens of US dollars for this test set). The base pipeline runs on a single CUDA GPU (embeddings) plus a remote vLLM endpoint (judge).

Ethics Statement

This work uses only the publicly released AraGenre shared-task data and introduces no new data collection; we did not gather, store, or infer personal information, and any user-generated text (e.g., social-media posts) originates from the official benchmark. In line with the shared-task rules, all predictions are produced automatically by the system: no submitted label was hand-edited and we did not reverse-engineer the hidden evaluation labels. We did read model outputs and the corresponding texts to design automatic correction rules, and the rule-precision estimates in §4 come from that inspection; this is author inspection of system outputs, not annotation of the evaluation set, and no such judgement enters a submission except through a deterministic rule.

Genre classification is a comparatively low-risk task, but two considerations warrant care. First, the taxonomy spans religiously and culturally sensitive categories (Islamic and Christian scripture, dialectal varieties, regional song lyrics), and we treat these descriptively as text-type/register labels; misclassification carries no endorsement or value judgement, and the system is not intended to assess the authenticity, orthodoxy, or quality of religious or cultural texts. Second, dialect-related labels can encode regional and social signals; models trained on skewed corpora may under-serve lower-resourced dialects (e.g., the near-noise-floor separation of Sudanese and Iraqi song lyrics we report), which practitioners should weigh before deployment.

Our correction stage sends Arabic text to third-party hosted LLM APIs; because the inputs are already-public shared-task data, this poses no additional privacy exposure, but the same pipeline should not be applied to private or sensitive text without review. The approach relies on large hosted models whose availability, cost, and behaviour may change, which limits equitable reproducibility; the base retriever-plus-open-judge pipeline is provided as a lower-cost, self-hostable alternative.

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A Annotation-Convention Divergence Between Splits

Five of the seven residual dev errors are texts that analyse religion from the outside, e.g. تشير بعض الدراسات إلى أن أساليب الوعظ تختلف بين المجتمعات ("some studies indicate that preaching styles differ between societies"). Our system labels these `Analytical Reports` (Informative); the dev gold label is `Religious Commentary` (Religious).

This is the *inverse* of the convention in the hidden test, where `islamic_book_description` ("texts summarising Islamic or religious books") is Informative and our consensus stage is rewarded for moving such texts out of Religious. Both conventions are stated in their own definition files, and the two files disagree about where writing *about* religion belongs. A system that follows the test convention is therefore necessarily penalised on dev, which bounds how far dev accuracy can be pushed without split-specific rules. It also supports our central claim: the convention is carried by the definitions, which is why supplying the full taxonomy, rather than the broad labels alone, changes model behaviour so sharply.

B Error Analysis Examples

Three error types dominate. (i) *Cross-family topic leaks*: a description *of* a religious book (يهدف هذا الكتاب إلى قراءة سيرة الرسول) is `islamic_book_description` (Informative) yet attracts Religious, while scripture misfiled as fiction (هؤلاء امراء بني عيسو, Genesis 36) moves the other way; consensus corrects both. (ii) *Attractor over-prediction*: long prose absorbed into `egyptian_song_lyrics`. (iii) *Ceiling classes*: for each of the five song-lyric classes, 87–99% of the texts we assign to it contain no marker from that dialect's cue list (`src/dialect_markers.py` in our repository ships the lists and reproduces the table; the rate is a property of the text, so it is measurable without gold).

C Negative Results

A *direct* six-way broad classifier regressed to 0.6206, confirming that deriving broad from the specific prediction beats classifying it directly; scoring all 74 candidates without retrieval pruning fell to 0.6517. For correction, plain self-consistency (0.7134), entropic optimal transport (0.7115), uniform-prior calibration (0.7124) and an unrestricted chain-of-thought pass (0.7131) all regressed below the 0.7139 base, whereas full-context consensus improved it. Cross-lingual rerankers failed (Aarab, 2026), regex rules added nothing the judge lacked, and the off-the-shelf dialect classifiers we tried did not expose the Iraqi/Sudanese distinction that NADI-style country-level tasks define (Abdul-Mageed et al., 2021).

D Full-Context vs. Broad-Only Prompting

Both consensus models were run over the same verification pool ($n = 2{,}635$) once with only the 6 broad definitions and once with all 74 specific definitions. Because the full-taxonomy prompt also carries an explicit type-vs-topic instruction, we added a third arm (broad definitions only *plus* that instruction) to separate the two factors; on the 1,403 book-description instances, GPT-5.6 calls 394 of them Religious with neither, 233 with the instruction alone, and 126 with the full taxonomy; `src/trap_table.py` rebuilds these three counts from cached verifier outputs with no API access. The table below reports the two original arms. Table 3 reports how often each model's verdict still agrees with the base system's broad family. Broad-only prompting collapses on families whose members are *about* another family's subject matter (Learning, Informative), which is exactly the topic trap; the full taxonomy repairs it. On the 1,403 book-description instances, joint Religious assignments fall from 235 (broad-only) to 37 (full taxonomy), an 84% reduction.

Base family	n	broad-only	full-taxonomy
Informative	1682	49%	**78%**
Learning	305	18%	**73%**
Interactive	153	46%	**73%**
Legal	281	53%	**55%**
Creative	151	50%	47%
Religious	63	41%	**46%**

Table 3: Share of instances where the verifier's broad verdict matches the base family, by prompt context. This measures *agreement with the base system*, not accuracy: the hidden test has no released gold, so we can show that full context changes verifier behaviour and that the change is consistent with the taxonomy's own conventions, but we cannot certify it as an accuracy gain. The dev contrast in §6.1 is the gold-anchored counterpart.

E Attractor Drain

The drain is deterministic: it fires when the base label is an attractor class and the chain-of-thought pass names a different sibling inside the same broad family. No score threshold or entropy criterion is used, and broad is held fixed by construction.

Class	base	moved
`literature_fiction_bk_desc.`	598	208 (35%)
`egyptian_song_lyrics`	979	212 (22%)
`msa_poetry`	608	52 (9%)
`culture_encyclopedia`	455	41 (9%)
`history_encyclopedia`	632	41 (6%)

Table 4: Attractor classes and how many predictions the drain reassigns to within-family siblings when applied to the base predictions alone. In the submitted pipeline consensus is applied first and the drain fires on 383 of these instances.

F Calibration

Marginal calibration rescales each candidate's score by a tempered estimate of its corpus-level prior, $\tilde{p}(y) \propto \hat{p}(y)^{\alpha}$ with α=0.5; α was chosen on the released training labels, where α=0.75 over-flattened once the label space grew from 7 to 74 classes. Rare-class correction uses the EM procedure of Saerens et al. (2002), which iteratively re-estimates the class priors of an unlabelled test set and reweights the classifier posteriors accordingly; we run it to convergence on the judge posteriors. Both are applied before the argmax and neither changes the broad family.

G Hyperparameters and Prompts

Retriever: Qwen3-Embedding-8B loaded in 4-bit NF4 with fp16 compute (the configuration the submitted run used; only the cosine ranking is consumed and NF4 preserves embedding direction), 4096-dim, L2-normalised, `max_seq_length` 512, the top 20 specific genres cached per text, retrieval k=12 narrowed to a family-restricted candidate set of at most 6. Judge: `gemma-4-31b` via vLLM, setwise 0–100 scoring, temperature 0, fixed seed, 3 retries with JSON-regex fallback, permutation self-consistency K=3 on confusable families, marginal calibration α=0.5 with SLD rare-class correction, embedding-rank guard at rank 3. Consensus verifiers: GPT-5.6 Luna (`reasoning_effort=low`, $\leq$500 completion tokens) and Gemini-3.6 Flash ($\leq$400 output tokens), both at temperature 0 with the full 74-definition taxonomy ($\approx$5.7K tokens) in the system prompt and the text truncated to 1500 characters. The judge prompt asks for a JSON array of integer scores over the candidate definitions; the consensus prompt asks for a single exact `specific_genre` identifier and states that a factual description *of* a book is a `*_book_description`, not the book's subject genre.

Throughput. A consensus call takes 2.0–2.8 s; at 24–32 workers the full GPT pass over 27,972 instances takes 35 min and the 8,041-instance Gemini pass 18 min.

H Official Submission Record

Every hidden-test number we report is an official evaluation of a submitted file, so the progression is traceable to scored submissions rather than to local approximations.

Two readings this record supports. Submission 867465 isolates the surface rules on top of 867323: they rewrite 32 labels (23 `quran`, 9 `hadith`) and leave the score at 0.6881 against 0.6882. The rules are precise but score-neutral at that point, so the 0.6882→0.7139 gain belongs to the judge and calibration changes, not to them. They were kept for a different job:

ID	System	Hier. F1
865908	genre-general zero-shot judge	0.6067
866538	+ embedding retrieval gate	0.6812
867323	family-restricted, derived broad	0.6882
867376	direct six-way broad judge	0.6206
867451	unpruned 74-way scoring	0.6517
867465	+ surface rules only	0.6881
870640	base system	0.7139
870841	uniform-prior calibration	0.7124
870979	self-consistency (K=3)	0.7134
870981	entropic optimal transport	0.7115
871127	chain-of-thought applied wholesale	0.7131
871609	+ 2-model consensus (pool)	0.7224
872511	+ safe broad expansion	0.7256
872515	**+ full consensus + drain**	**0.7352**

Table 5: Official submissions behind Table 1 and Appendix C. Submission 872515 is the final system: 27,972 predictions over 27,972 gold instances, none missing. Submissions were made sequentially over four days, so the progression is a record of scored configurations rather than a single controlled sweep.

in the final pipeline they fire on 1,087 instances and agree with the base label on every one, so their whole net effect is to restore 75 labels that consensus or the drain had moved away (56 `youtube_comments`, 19 `quran`). They guard the correction stages rather than adding score. Submission 871127 applies the chain-of-thought pass to every instance and scores 0.7131, below the 0.7139 base; the same pass restricted to attractor classes is what the drain uses. Chain-of-thought helps here only where it is aimed.

Orthographic signals. Share of each group carrying heavy diacritisation (a mark on over 10% of its Arabic characters) or at least one emoji, over the final predictions: `quran` 77%/0% (n=691), poetry 43%/0% (n=1,606), `hadith` 1%/1% (n=647), book descriptions 1%/0% (n=3,916), Interactive 0%/47% (n=1,103). Both are properties of the text, so no gold is needed; `src/orthographic_signals.py` reproduces the table.

Software. `torch` ≥2.1, `transformers` ≥4.44 (`https://github.com/huggingface/transformers`), `sentence-transformers` ≥3.0 (`https://github.com/UKPLab/sentence-transformers`), `bitsandbytes` ≥0.43 (`https://github.com/bitsandbytes-foundation/bitsandbytes`), vLLM (`https://github.com/vllm-project/vllm`), `openai` ≥1.99 (`https://github.com/openai/openai-python`) and `google-genai` ≥0.3 (`https://github.com/googleapis/python-genai`). Model weights: Qwen3-Embedding-8B (`https://huggingface.co/Qwen/Qwen3-Embedding-8B`).